Skills Summary

Terms, Factors, Coefficients and Degree

Use this reference while completing your worksheet or when studying.

Skill 1 — Terms, coefficients and degree

The vocabulary, on one polynomial: $3x^4 - x^2 + 5x - 2$

- **Terms** are the pieces joined by $+$ and $-$; the sign belongs to the term.
- The **coefficient** is the number multiplying the power. A lone x has coefficient 1; a lone number has degree 0.
- The **degree** of the polynomial is the largest exponent that appears.
- Handy check: $P(1)$ is the sum of all the coefficients.

Example: For $P(x) = 3x^4 - x^2 + 5x - 2$, list the coefficients and find $P(1)$.
Step 1. The coefficients are what I want, and substituting 1 turns every power of x into 1, so the sum of the coefficients is exactly the value $P(1)$.
Step 2. Coefficients in order: 3, -1, 5, -2 (degree 4, and the constant has degree 0).
Step 3. $P(1) = 3 - 1 + 5 - 2$
$\Rightarrow \quad P(1) = \mathbf{5}$

Try it: For $R(x) = 2x^3 + 4x^2 - x + 6$, list the coefficients and find $R(1)$.

check: **11**

Skill 2 — From a product to a sum (area model)

Rule: every cell of the rectangle is one partial product, and the total area is their sum.

$$(x+p)(x+q) \;=\; x^2 + (p+q)\,x + pq$$

The middle coefficient is the *sum* of the two numbers; the constant is their *product*.

Example: Multiply $(x+5)(x+2)$.
Step 1. Two binomials multiplied means four partial products, so I draw the two-by-two area model rather than guessing the middle term.
Step 2. Cells: $x \cdot x = x^2$, $x \cdot 5 = 5x$, $2 \cdot x = 2x$, $2 \cdot 5 = 10$.
Step 3. Add, combining the like terms $5x$ and $2x$: $x^2 + 7x + 10$.
$\Rightarrow \quad \boldsymbol{x^2 + 7x + 10}$

Try it: Multiply $(x+3)(x+4)$.

check: $\boldsymbol{x^2 + 7x + 12}$

Skill 3 — From a sum back to a product

Always look for a common factor first: pull out the largest number that divides every coefficient and the lowest power of x that appears in every term. **Then, for x^2+bx+c:** find two numbers with product c and sum b; they are the constants of the two binomial factors.

Example: Factor $10x^3 + 15x^2$, then factor $x^2 - 2x - 15$.
Step 1. The first has a common factor in every term, so that is a common-factor job; the second has no common factor but is a trinomial, so it is a product-and-sum job.
Step 2. 10 and 15 share 5, and both terms carry x^2: $10x^3 + 15x^2 = 5x^2(2x + 3)$.
Step 3. For $x^2 - 2x - 15$: two numbers with product -15 and sum -2 are -5 and 3.
$\Rightarrow$ $\boldsymbol{5x^2(2x+3)}$ and $\boldsymbol{(x-5)(x+3)}$

Try it: Factor $x^2 - 4x - 21$.

check: $\boldsymbol{(x-7)(x+3)}$

(!) Watch out: the degree of a *product* is the sum of the degrees of its factors, so $(3x^2 - 1)(x+5)$ has degree 3 — you never have to expand to find it. And a missing number is never a missing coefficient: in $x^2 + x$, the coefficient of x is 1.